



Welcome to MSU CS

Getting Started in Computer Science

Learning Computer Science

Starting computer science classes can feel exciting and challenging, especially if you are new to programming or still building confidence using your computer and technical tools. You are not expected to come in with prior experience, but it is important to use the various supports here at MSU to help you succeed.

Learning computer science is not about already knowing how to code — it is about being willing to think, experiment, ask questions, make mistakes, learn, and try again. That is exactly how real work happens in industry.

[How to Embrace Failure | Reshma Saujani](#) "If you fail fast, fail often, and fail quickly you will learn to exercise your bravery muscle leading to a much more joyful life."



Introductions

Sit with someone using same operating system (Mac or Windows).

Introduce yourselves by sharing the following and rotate to next person

1. Your name and pronouns
2. One statement about their experience with technology (true or false)
Others guess whether it's true.

Then we will rotate and share

3. How do you feel about starting CS 1050?
4. What are you excited and nervous about in programming?

Contents

[Metro Computer and Software \(Apps\)](#)

[Operating System Fundamentals](#)

[Version Control](#)

[Integrated Development Environment](#)



[Struggling with technology ?](#)

Remember we must keep a sense of humor. We all make mistakes when learning.

Computer and Software (Apps)

Computer/Laptop Recommendation [Student Resources](#)

1. Recommended: at least 16 GB but 32 GB is best or more.
2. Recommended: CPU (single or multi-core) with a main frequency >2.5GHz (e.g. ARM-based Apple M1, M2, M3 chips are okay).
3. NOT Recommended: Chromebooks with low frequency CPU, Tablets, iPads.
4. Both MacOS and Windows OS are fine (with VirtualBox or VM).

[Article - NetID General Information](#) Your MSU Denver NetID is the first part of your MSU Denver email address, before the @. All students at MSU Denver are provided with a NetID and password when they join the university. Your NetID permits secure access to a variety of applications and services, including:

- [How do I access my OneDrive on the web?](#)
- [How do I download Office 365 applications on my personal devices?](#)
- [Canvas Quick Start Guide for Students](#)
- [Wireless printing in student labs](#)
- [What are MSU Denver's Wi-Fi networks?](#)
- On-Campus Computer Labs:
 - [Map & Hours of Computer Labs](#)
 - [Auraria Library Computers](#)



Get connected to the wifi and access your onedrive on the web. Help others if you already know.

Operating System Fundamentals

Mac and Windows

Getting Organized

Organization is important as a student and in a career.

Create a folder for your work on your computer or in google drive with the course title.

Set up your browser to ask where download files instead of automatically downloading in the downloads folder.

Search for instructions on how to do this or see next slide to set up using Google Chrome

How messy is your desktop? What is in your downloads folder?

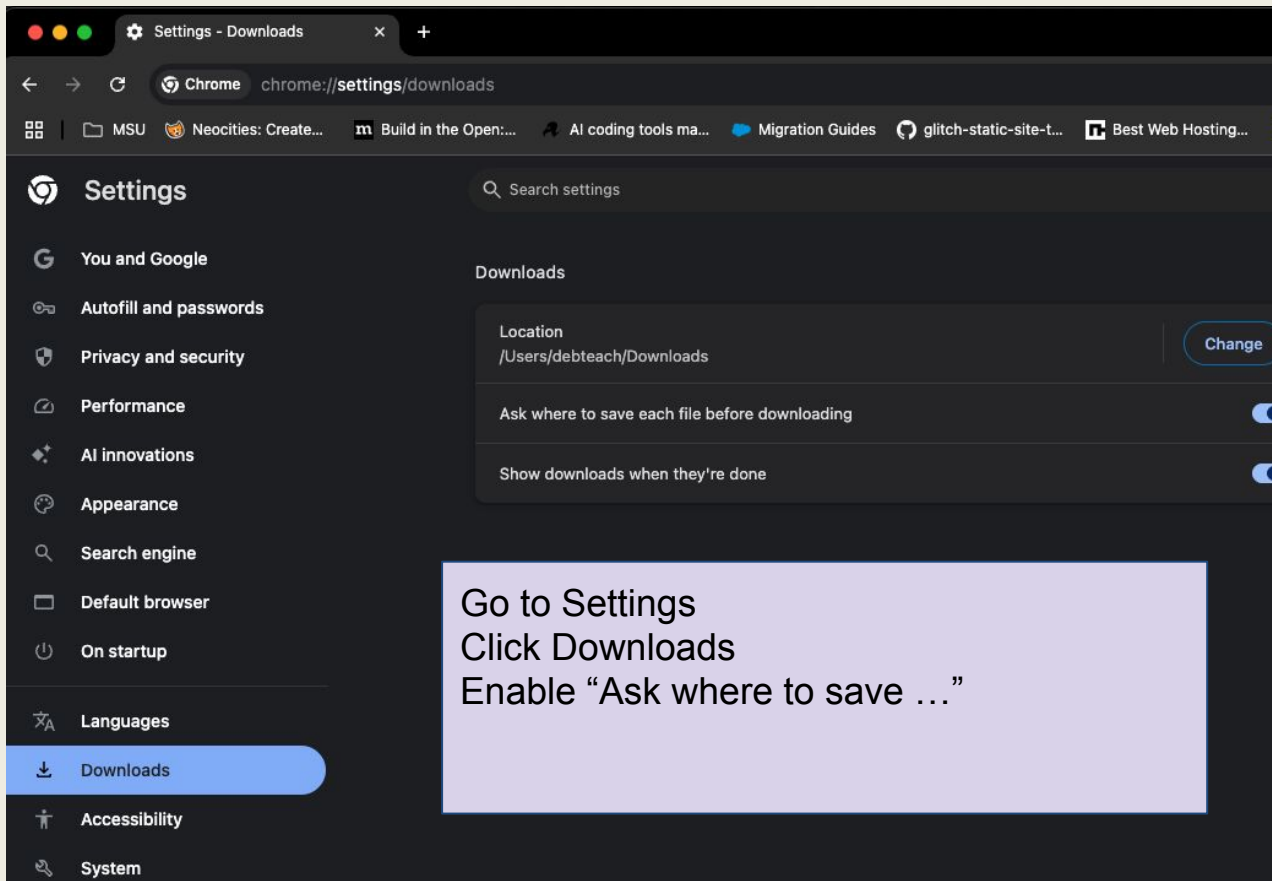


Where do you files go when you download from your browser?



There are too many files in these folders
Please remove any files you don't need anymore.

Browser Download: Ask Where to Save Files



For example you can
[Download Google Chrome Web Browser](#)



Navigating Folders and File Paths

Eventually you will navigate your computer using command line in future classes.

For now it is good to understand the folder locations and types of files.

You should see the folder path to your files.

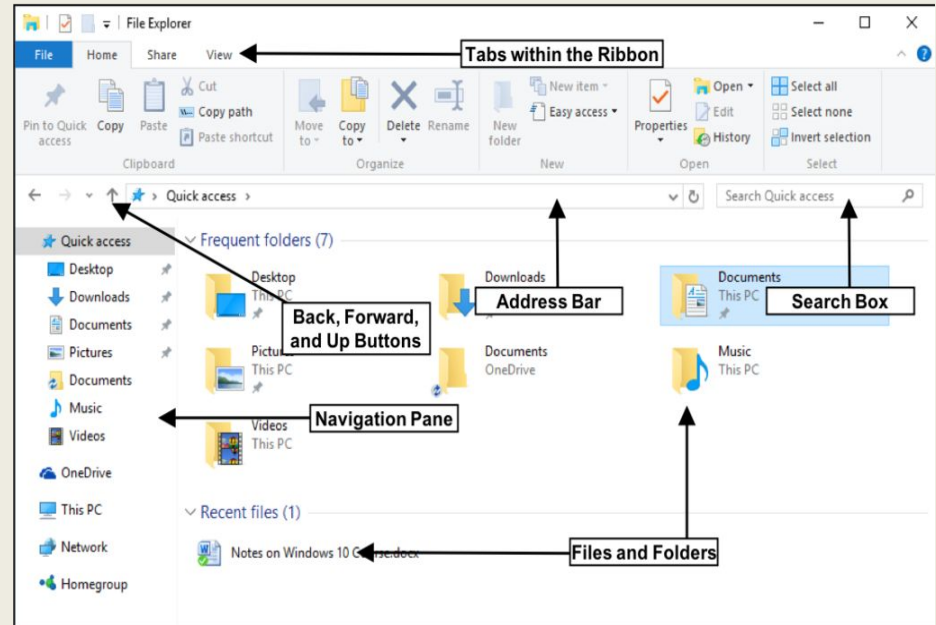
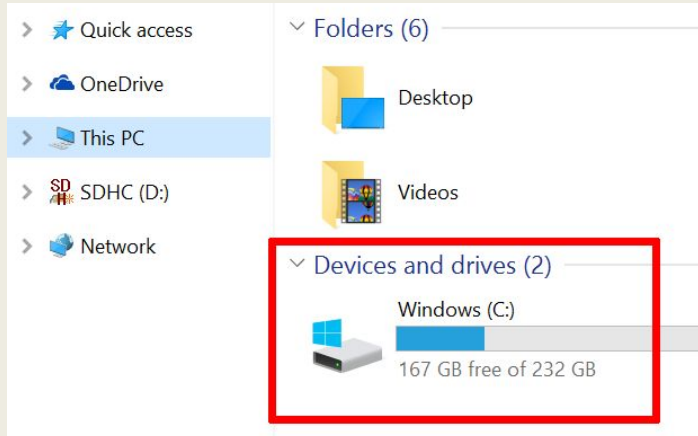
You should see the file types and hidden files on your computer.

Windows Folder Structure

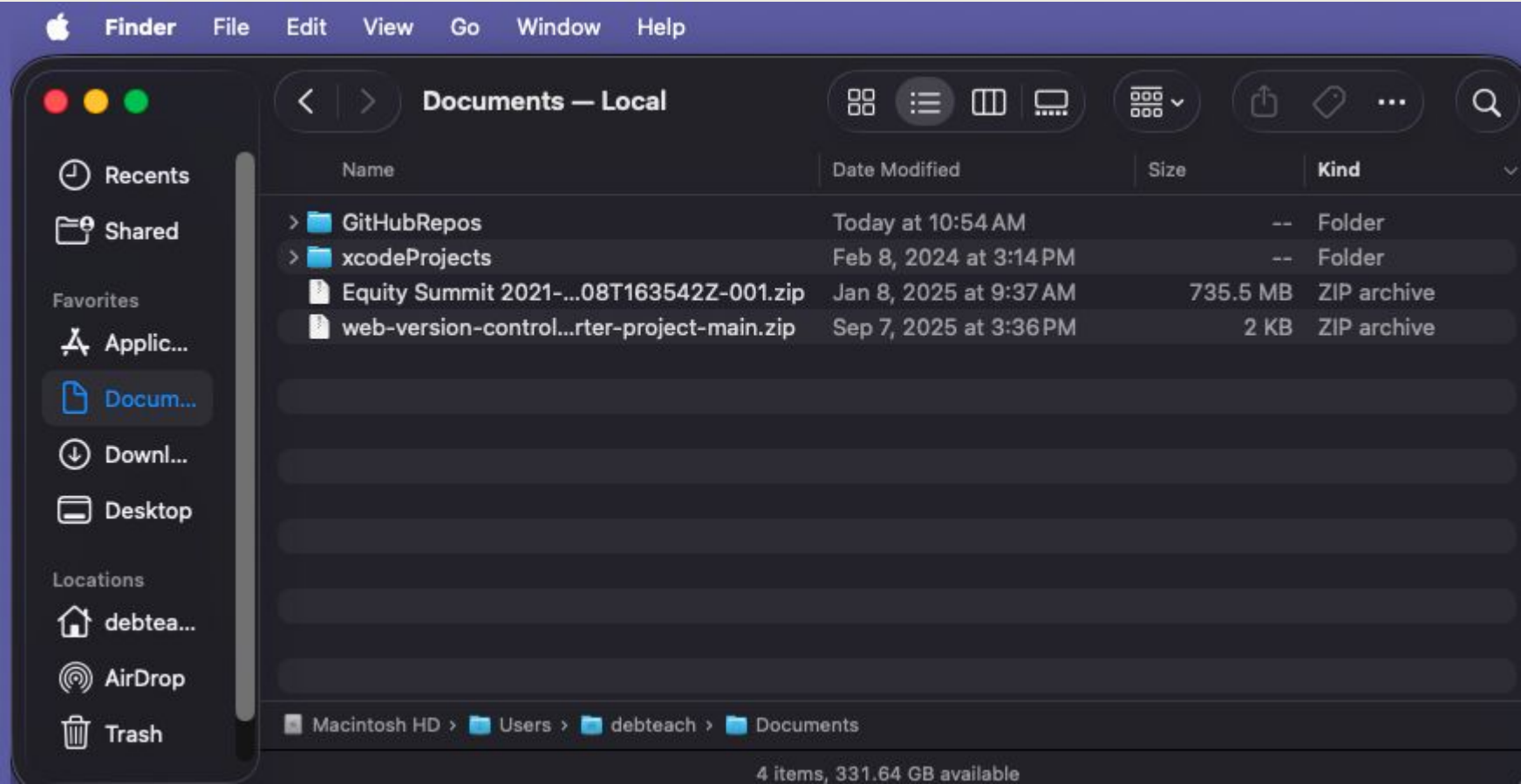
[File Explorer in Windows - Microsoft Support](#)

The C: Drive - Think of it as: The “main storage area” of the computer.

- On Windows, C: is the main hard drive.
- It contains the Windows operating system, programs, and user files.



Mac Folder Structure



Windows Folder Paths and File

[How to Always Show the Taskbar in Windows 10 and 11](#)

[Turn On or Off Display Full Path in Title Bar of File Explorer in Windows 11](#)

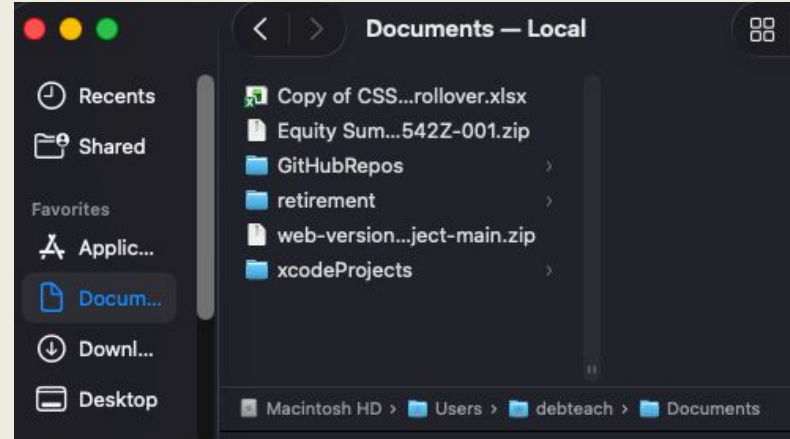
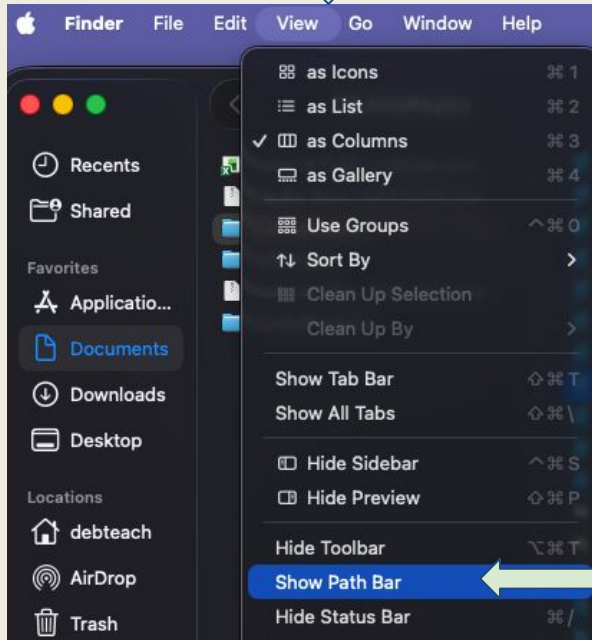
[How To Show Hidden Files And Folders In Windows](#)

[How to Make Windows Show File Extensions](#)

Mac Folder Paths and Files

Turn on your Toolbar, Path Bar and Status Bar

1. Go to finder and click view menu

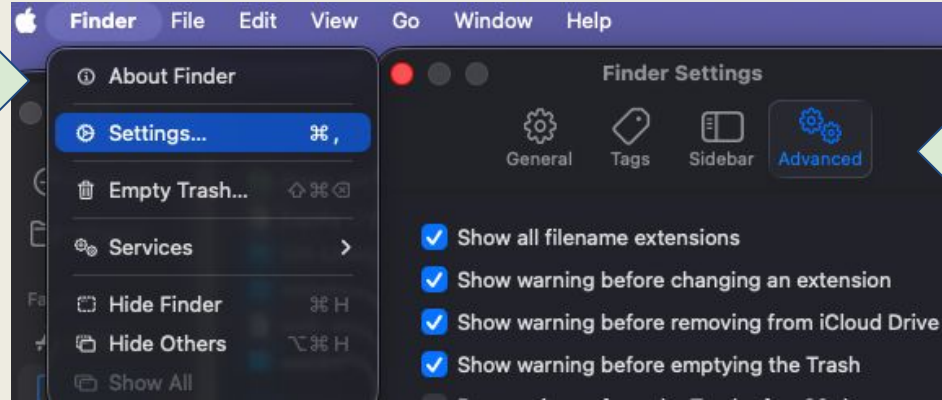


Now you can see the path to the folders and files on your computer.

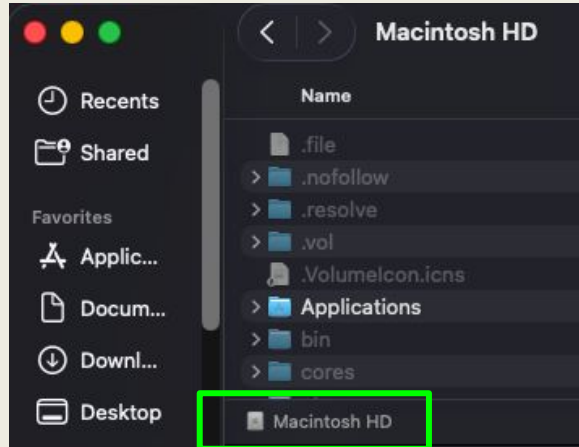
2. If any say Show click them to turn on

Mac Show File Extensions and Hidden Files

1. Go to finder
click view menu
click settings



2. Click advanced and
check options.



Mac: To reveal hidden files in Finder
go to Go to Macintosh HD and press Shift + Command + . (period).
[Show Hidden Files and Folders on a Mac Using Finder](#)

Zip Files

A ZIP file is a compressed folder that contains one or more files.

Think of it like a packed suitcase—you must unpack it before using what's inside.

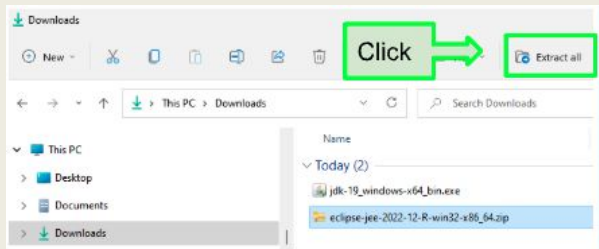
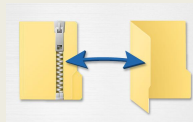
Why software comes as ZIP files

- Makes download sizes smaller
- Groups multiple files into one package
- Works on all operating systems

Trying to run programs from inside the ZIP without extracting causes missing file errors.

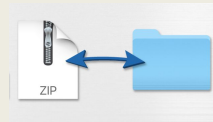
How to open a ZIP file Windows

1. Right-click the ZIP file
2. Select Extract All...
Choose a folder → Extract



How to open a ZIP Mac

1. Double-click the ZIP file
2. macOS automatically extracts it into a normal folder



Windows exe File

An EXE (executable) is a program you run directly on Windows. Think of it like A “Start Installation” button packaged as a file.

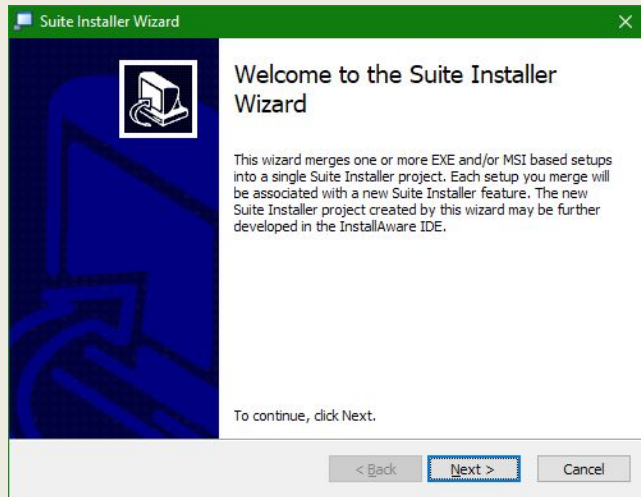
What EXE files do

- Install software (most common)
- Run applications
- Launch setup wizards

How to use an EXE file (safe version)

1. Download the EXE
2. Locate it in Downloads
3. Double-click
4. Follow the setup steps (Next → Next → Finish)

Important safety rule: Only run EXE files from trusted sources (e.g., GitHub Desktop, Eclipse.org, Oracle, Adoptium).



Mac DMG Files

A DMG file is a macOS Disk Image—a virtual “disk” that opens on the desktop. Think of it as a temporary virtual USB drive that contains an app.

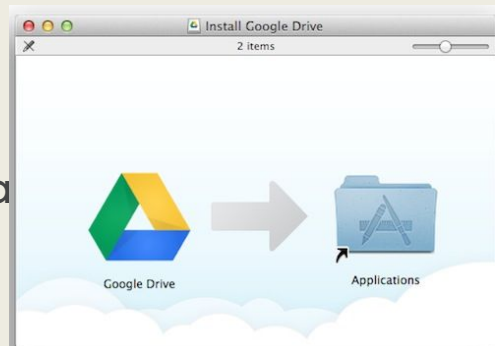


How DMG installation works

1. Double-click the .dmg
2. It opens a window that usually shows the app on the left and the Applications folder on the right
3. Drag the app → Applications
4. Eject the DMG (like removing a USB drive)
5. Move or delete the DMG file from Downloads

Why apps use DMG files

- Makes it easy for users to drag the app into Applications
- Keeps the installer simple (no setup wizard)



Determine Operating System and Architecture

Operating system (OS) software that runs on your computer to manage the computer's basic functions. In computer science you will need to install different tools on your computer.

When downloading programming tools, installers often ask you to choose between AArch64, x86_64, ARM, Intel, or riscv64.

These terms describe the CPU architecture, the type of processor inside a computer. Software must match the architecture or it won't run correctly.

- On Windows -> Settings -> System -> About -> Under System type:



- 64-bit operating system, x64-based processor is x86_64
- ARM-based processor is AArch64

- On macOS Apple menu -> About This Mac



- Intel is x86_64
- M1 / M2 / M3 is AArch64 (Apple Silicon)

Processor 2.6 GHz Quad-Core Intel Core i7

Chip Apple M1 Pro

- Most linux are using x86_64



Add to Toolbar for Easy Access to Software (Apps)

After you install software you can add easy access to opening it by adding it to your desktop toolbar. This way even if you close the app it will stay in the toolbar

Mac: After you open the software you installed

1. Right click the icon in the toolbar
2. Click **Options**
3. Select **Keep in Dock**

Windows:

[How to Add a Desktop Shortcut and Pin Apps to the Taskbar in Windows 10 & 11](#)



Set Up

Get set up with the following or help someone if you have it set up:

1. Log into your portal
2. Check your email
3. [Canvas Quick Start Guide for Students](#)
4. [Wireless printing in student labs](#)
5. [What are MSU Denver's Wi-Fi networks?](#)

Scavenger Hunt

Getting Organized now will save time later.

Work together to complete these tasks

1. Open your portal and find your schedule.
 - a. Find what room you are in for your CS1050 class
2. Log into canvas.
3. Access your MSU One Drive
4. Go to your email and send Deb and ThienNgo and email your CS1050
5. Use OneDrive or Google Drive
 - a. Create A folder for Spring 2026 semester and in that put a folder for CS1050
 - b. Create a document called important information. Add 2 or 3 items you learned so far that you may need to access later. For example maybe the link to these slides or link to onedrive.
6. Download these slides and put in your folder.
7. Go to the second floor and take pictures of
 - a. Your room, the CS student lounge, the CS department front desk, your CS instructors office
8. Email Deb Harding and ThienNgo when you finish



Version Control

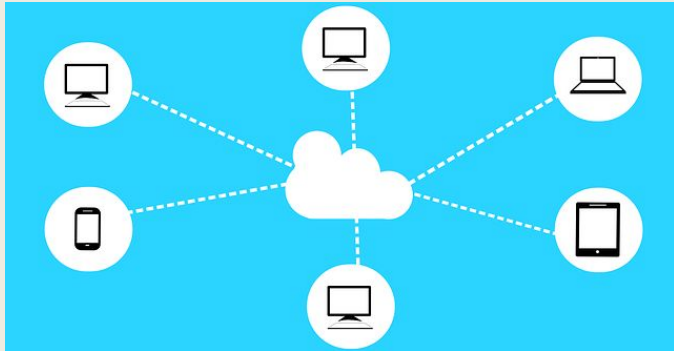
Git and Github



Safety First: Back Up Securely

Computer devices can fail, data can be corrupted, computer devices can be lost, stolen, or destroyed. It is important back-up (make more than one copy) and secure your data.

Many industries have secure cloud storage for multiple employees to contribute to projects. They can work on local computer and back up to a cloud server and integrate their work with others to track changes and versions.



[Image: Pixabay](#)

Open a word or google doc. View the version history.

- [View previous versions of Office files - Microsoft Support](#)
- [How to Check Google Doc History: See Changes & Restore Edits](#)

You will be setting up version control for your computer science projects.

What is Version Control?

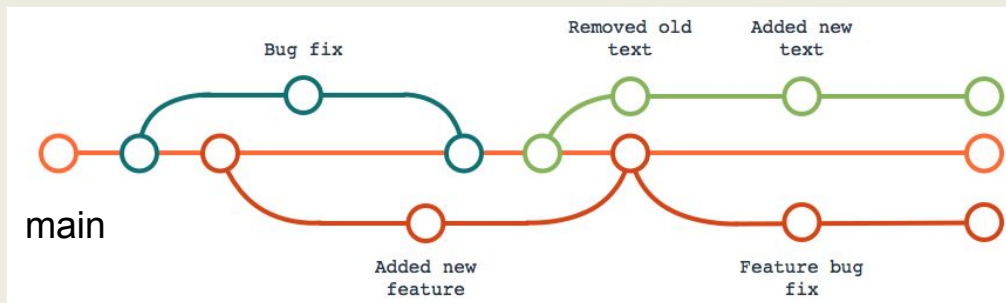
Think of version control as a time machine that keeps snapshots of your code.

- Tracks changes in files over time.
- Helps collaborate efficiently without overwriting others' work.
- Supports reverting to previous versions when needed.

Managing Versions

- As time goes on, there will be more than one version of the software
- Versions of the software need to be separated somehow in the version control system
- Version control software stores ALL code and ALL changes

You will be using [Git](#) as the version control tool in computer science courses.



Overview How GitHub Works

You will not be using all the industry features of git and Github in the beginning CS courses. You will eventually be using the various features in future classes.

Here is an overview on [How Github Works](#)

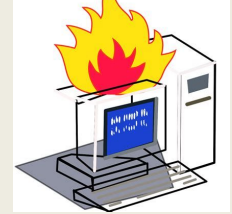


Backing Up Data (GitHub) and Version Control (Git)

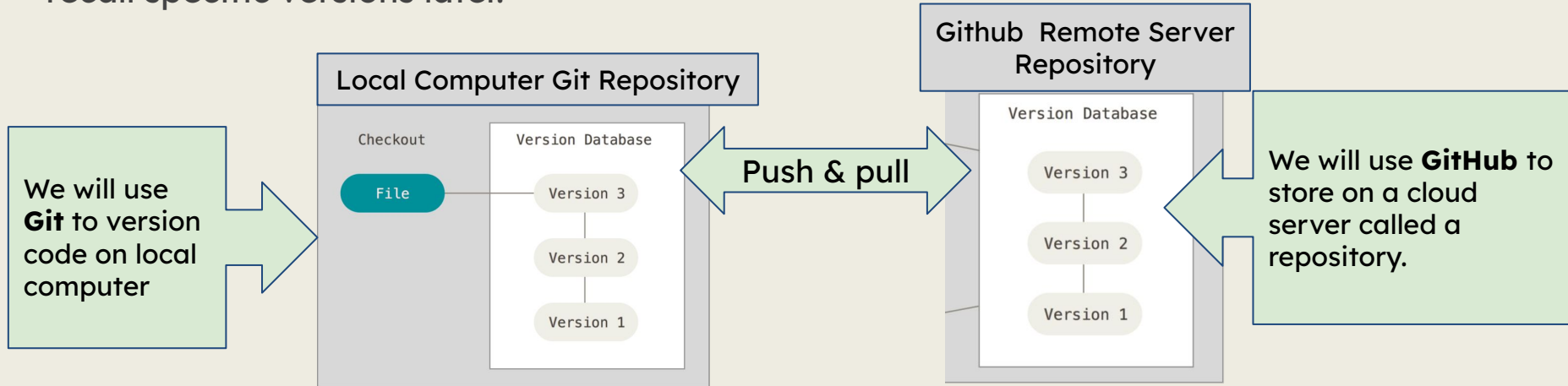
You will back-up (make more than one copy) and secure your data using GitHub



Not a valid reasons for
losing your work



We will use **Git** version control to records changes to your files over time so that you can recall specific versions later.



Git Version Control and Github Server

Git is a **version control tool** on your computer that tracks every change you make in your local repository. Git helps you:

- save a “snapshot” of your project (called a **commit**)
- track what changed and when
- keep a local history of your work
- experiment without fear of losing your original code

GitHub is a **website that stores your Git repositories online in the cloud**. GitHub lets you:

- back up your work in the cloud
- access your code from any computer
- share projects with your instructor
- store different versions in one place
- include your projects in a portfolio



[Git Made Easy: Learn Version Control & Collaboration in Minutes!](#)

Git is the tool to version, GitHub is the online home for your projects.

Git Local Repository and Github Remote Repository

A Git repository (or repo) is a storage space where your project files live, Think of it as a folder where you track every change you make to your files. There are two main types of repositories:

- Local Repository: Stored on your own computer; contains your files and a hidden .git folder.
- Remote Repository: Hosted online (e.g., GitHub) so you can back up and share your work.



How Git and GitHub Work Together

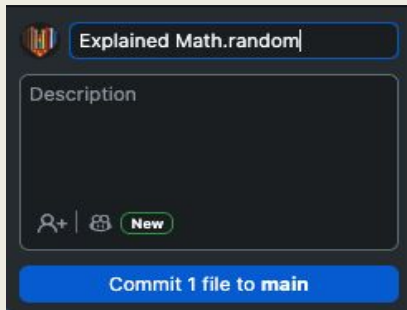
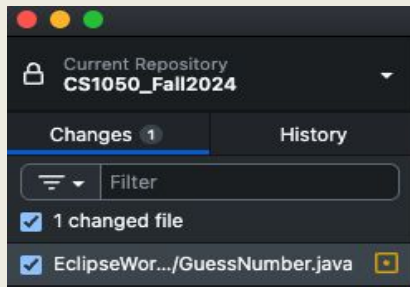
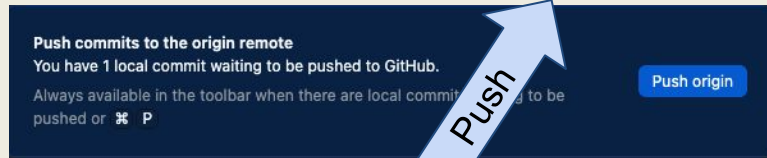
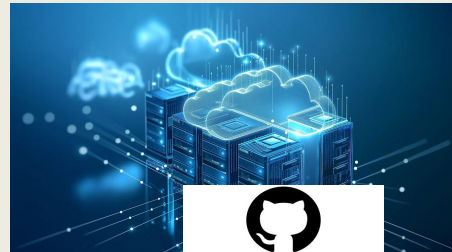
Track changes using git and back up your versioned code on GitHub

A commit saves a version of your project that includes

- The changes you made (tracked by Git)
- A message (comment) explaining what you did

A push sends your commits to GitHub so:

- Your work is backed up online
- You can view changes on GitHub Pages



Commit with
message comment
to version locally

Changes on Local
computer



Version Changed Code: Commit and Push

Track changes on your local computer using git

A commit is a saved version of your project.

Each commit includes:

- The changes you made (tracked by Git)
- A message explaining what you did



Comment
Message Title

More Details

Click commit to make a new version
on your local computer.

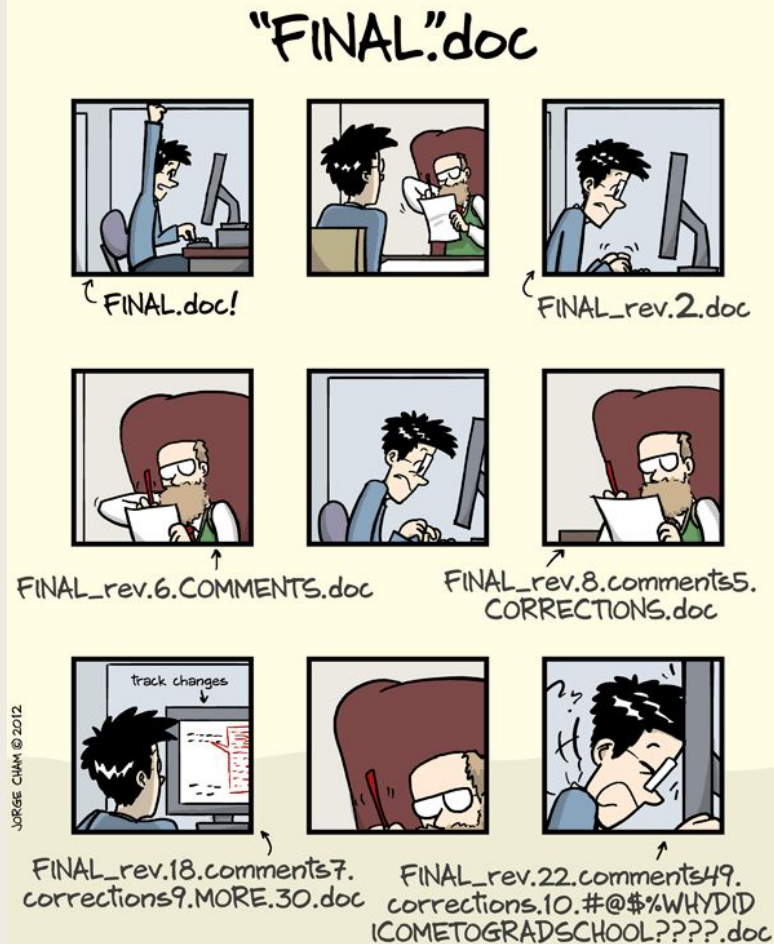
A screenshot of an IDE window. The top bar shows 'Current Repository: CS1050_Fall2024', 'Current Branch: main', and a 'Fetch origin' button. The left sidebar has a 'Changes' tab with '3 changed files' listed: 'EclipseWorkS.../GE01Library.java' (checked), 'Ecli.../PlaylistIteration02Test.java' (unchecked), and 'Eclips.../Ridelteration01Test.java' (unchecked). The main editor shows the code for 'GE01Library.java'. The code includes imports, comments, and a class definition for 'LastNameFirstName_M01Library' with a 'Book' class inside. At the bottom, there is a commit interface with a text field containing 'GE01LibraryCode', a description 'Implemented encapsulation', and a 'Commit 1 file to main' button.

Commit Message

You've been putting nice descriptive messages each time you committed code into your version control system, right?

Here's where they matter. Just as each commit gets a revision number, your version control software also keeps your commit messages associated with that revision number, and you can view them.

You and your team needs to know what was changed.



Show Local Rep and View Github Repo

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.

Open the repository in your external editor

Select your editor in [Settings](#)

Repository menu or   A

Open in Xcode

View the files of your repository in Finder

Repository menu or   F

Show in Finder

Open the repository page on GitHub in your browser

Repository menu or   G

View on GitHub

GitHub Desktop

Show: Open local repo to go to files

View on Github Server

View: Open github repo to go to updated file on

Name	Date Modified	Size	Kind
CS1050Mod01_02	Today at 4:00 PM	--	Folder
.classpath	Feb 2, 2025 at 3:06 PM	380 bytes	Document
.project	Feb 2, 2025 at 3:06 PM	373 bytes	Document
.settings	Feb 2, 2025 at 3:06 PM	--	Folder
GuessNumber.class	Today at 2:58 PM	2 KB	Java class file
GuessNumber.java	Today at 2:58 PM	831 bytes	Java source code
CS1050Mod3	Yesterday at 4:03 PM	--	Folder
CS1050Mod4	Today at 9:46 AM	--	Folder

Macintosh HD > Users > debmhteach > Documents > GitHubRepos > CS1050_Fall2024

Files

main

Go to file

EclipseWorkspace

CS1050AssingSol

CS1050M02

CS1050Mod01_02

CS1050_Fall2024 / EclipseWorkspace / CS1050Mod01_02

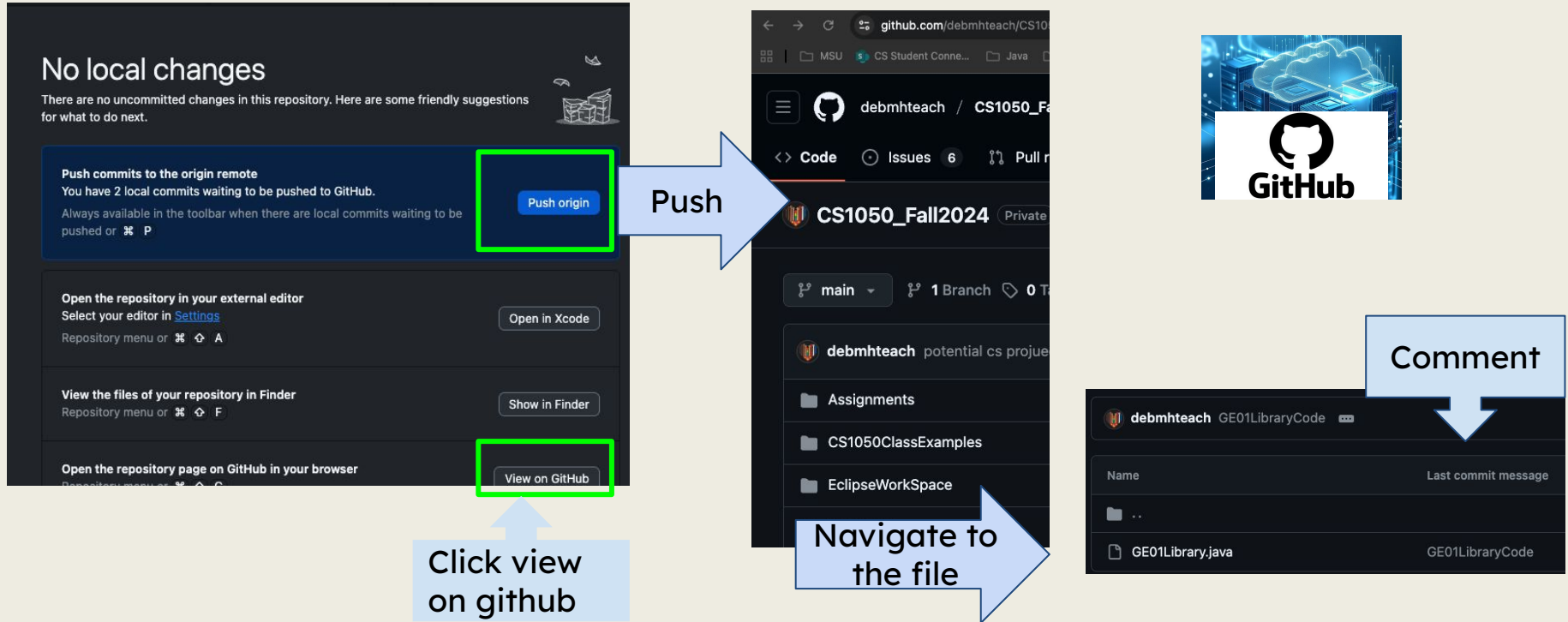
debmhteach Explained Math.random 042435b · 47 minutes ago History

Name	Last commit message	Last commit date
..		
GuessNumber.java	Explained Math.random	47 minutes ago

Version Changed Code: Commit and Push

Back up your versioned code on GitHub

A push sends your commits to GitHub so your work is backed up online



Version Control: Fetch and Pull

Update Local Repository with GitHub Server Changes

Fetch checks the GitHub cloud (remote) for updates to the repository.

- It does not bring those changes into your local files yet — think of it as checking for updates but not installing them.
- You use Fetch when you want to see if anything has changed on GitHub before deciding to update your local version.

A pull brings the changes from the GitHub remote repository into your local repository.

It is the combination of Fetch + Merge. This means it includes changes into your local files.

- You use Pull after a fetch to download and apply changes made by you (on a different computer)

Setting up a GitHub Repository

Ways to create a repository

- Create on GitHub
- Create Locally with Git

You will create a repository for this class using Github to store your files - Assignment documents, Class and Assignment code and technical documentation.

.gitignore File - What files to ignore that might be in your development environment. For example, I ignore macOS system files, java project files, java class files, etc

README file - Documents information about the project

Setting Up Git and Github Desktop for Java

Follow these directions to set up [Java Git and Github Desktop Version Control](#)

Git and GitHub Tips and Resources

Remember

- Commit and Push often
- Version code and assignments

? Having issues with GitHub Desktop?

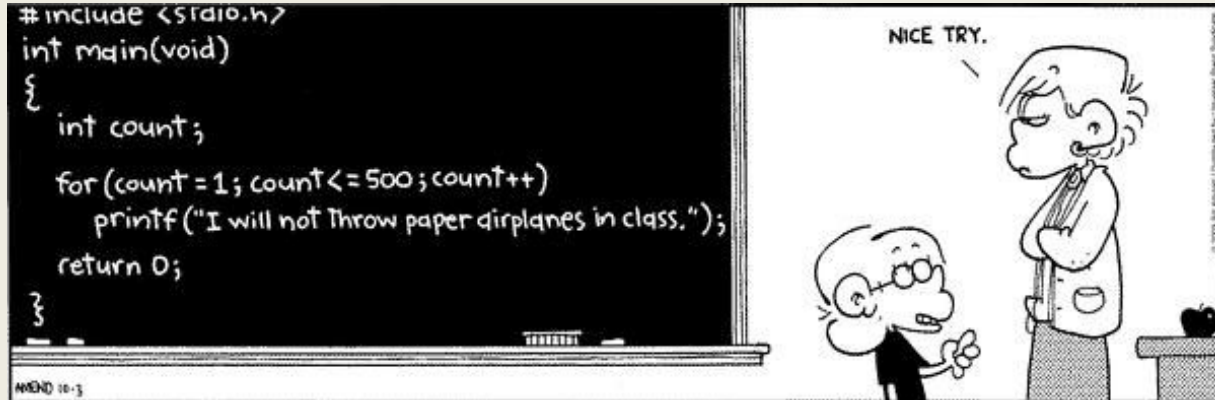
- My local changes aren't showing up on GitHub? Commit and push?
- Updates on GitHub.com, how do I get that version locally? Fetch + Pull

Resources

- [CS-2-Community-HUB/github_desktop_guide.md](#)
- [GitHub Desktop documentation](#)
- If you are more experienced you can use command line. Connect with me if you want to do it this way.

Integrated Development Environment

Eclipse



Integrated Development Environment (IDE)

What is an IDE?

An Integrated Development Environment (IDE) is a software application that helps you write, edit, and manage code — all in one place.

Think of it as a toolbox for coding that combines:

- A text editor (to write your code)
- A preview or terminal window (to test or run your code)
- Helpful tools like:
 - Syntax highlighting (color-coded tags and text)
 - Autocomplete (suggests code as you type)
 - Error checking (shows when something is wrong)
 - Debugging (Step through code to understand what is happening or find errors)

Why Use and IDE?

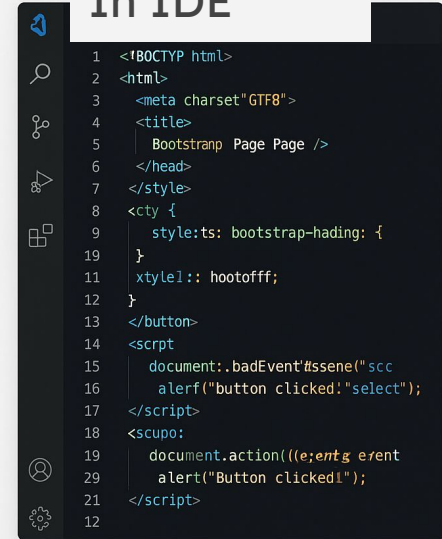
- Makes coding easier and faster
- Helps you avoid mistakes
- Organizes your project files in one place
- Gives you immediate feedback while you work
- Had debugger options to trace code and view information stored in variables.

Plain text

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Bootstrap Page/>
  <link>
    <lihref<csscloudflare.com/
css/stylesheetsstrap>
  </style>
</html>
<css {
  h1 class="bootstrap-heading-{
    Hello, Bootstrap!
  </button>
</script>
<document.addEventListener()
  alert("Button clicked!")
</script>
```

In VS Code

In IDE



AI Generated

Different IDEs can be used for different languages and use cases. Free IDE examples you might use.

- [VS Code](#) or [VS Codium](#) are a popular IDEs used for web and python development.
- Eclipse is a popular IDE used for Java Development ([Eclipse Setup for Java](#))



Debugging

When you find errors in the code, these are called **bugs**. Removing the bugs meaning fix the errors is called **debugging**.



[Grace Hopper found the problem, a moth in the computer.](#)

Different IDEs Used

Different IDEs can be used for different languages and use cases. For Java programming will use Eclipse.

Why and IDE?

- Makes coding easier and faster
- Helps you avoid mistakes
- Organizes your project files in one place
- Gives you immediate feedback while you work

Why Eclipse?

- Free & Reliable — No cost, works across Windows, macOS, and Linux.
- Industry-Standard Tool — Still widely used in Java development.
- Java-Focused Features — Offers auto-completion, auto-formatting, debugging, and more to make coding smoother.
- Real-World Learning — Helps you practice organization, testing, and debugging in a professional environment.

[10 Best Java IDE For Developers in 2025 - GeeksforGeeks](#)

Eclipse Set UP

Follow these directions to set up [Setup Eclipse](#)

Then practice using [Eclipse Cheat Sheet](#) and GitHub desktop

Wrapping Up

- How do you feel about starting CS 1050?
- What are you excited and nervous about in programming?
- What tips do you have? What questions do you have?

Other items you may want to set up or review

- [How do I download Office 365 applications on my personal devices?](#)
- [Canvas Quick Start Guide for Students](#)
- [Wireless printing in student labs](#)